SEG3101 Deliverable #2

Interview, Domain, Goals, and Interface

SEG3101 Fall 2024
Instructor: Wassim El-Ahmar

Group No. 24

Members:
Shabrina Sharmin 300230297
Amadou Kone 300146259
Hazzem Sukar 300286631
Alperen Akin 300290090

1. Introduction 3
Glossary / List of Acronyms 3
2. Domain Description 3
3. Initial Problem Description 5
3.1. Stakeholders and Their Goals 5
3.1.1. At least 5 important stakeholders and their goals 5
3.2. Scope of the System 5
3.2.1. General features within the scope of the system, and explicit description of what is not in the scope of the system 5
3.2.2. Description of dependencies and anticipated interactions with external systems 6
3.3. Main User Stories 6
User Story 1: Elicit Requirements from Virtual Stakeholders 6
User Story 2: Generate Domain-Specific Information 7
User Story 3: Create Personas Automatically 7
User Story 4: Validate Requirements 7
User Story 5: Generate User Stories 7
User Story 6: Produce Non-Functional Requirements 7
User Story 7: Simulate Stakeholder Meetings 7
User Story 8: Handle Ambiguities 7
User Story 9: Generate Use Case Scenarios 7
User Story 10 (Negative User Story): Handle Inaccurate Data 7
3.4. Preliminary Requirements 8
3.5. Major Risks and Their Potential Consequences 12
Risk 3: Data Privacy and Security Concerns 13
4. Interface Prototypes 13
Screen 1: Home Screen Dashboard 13
Targeted Stakeholder Goals: 14
Features: 14
Screen 2: Artifacts Management Interface 15
Targeted Stakeholder Goals: 15
Features: 15
Screen 3: Task and Agenda Management Interface 16
Targeted Stakeholder Goals: 16
Features: 16
5. Team Description and Roles 17
6. References 17

1. Introduction

This Software Requirements Specification (SRS) document outlines the requirements for the development of a Virtual Stakeholder System (VSS). The primary objective of the VSS is to address the challenges faced during requirements elicitation sessions when key stakeholders are absent or hard to reach. By leveraging advanced technologies such as large language models (LLMs), the VSS aims to facilitate progress in requirements engineering, gather information in unfamiliar domains, generate essential artifacts, and validate requirements. This document will serve as a comprehensive guide for the development team, providing detailed information about the domain, stakeholders, system scope, user stories, requirements, risks, and interface prototypes.

Glossary / List of Acronyms

- VSS: Virtual Stakeholder System
- SRS: Software Requirements Specification
- LLM: Large Language Model
- UML: Unified Modeling Language

2. Domain Description

The VSS is an advanced tool that is designed to assist the requirements engineers and other team members such as software developers, managers, business analysts who contribute to the project at hand. The main objectives of VSS are to simulate the presence and need of stakeholders to make the elicitation process easier and effectively make progress in the requirements engineering phase. The program leverages advance language models and generates data driven domain specific insights, feedback, user stories, refinements. Some of the primary goals of VSS are as follows:

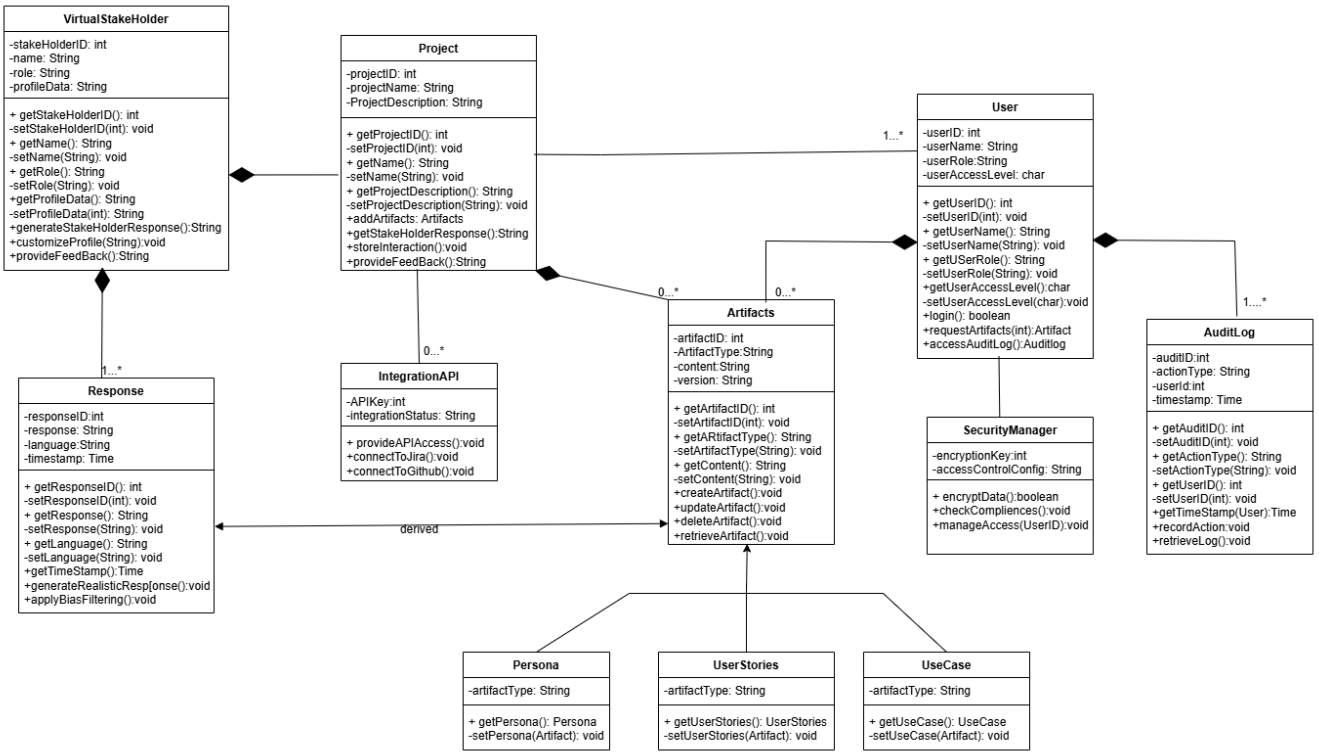
- Imitate a stakeholder who has the necessary domain specific knowledge and expertise
- Generate data driven feedback, responses, requirements, artifacts based on the domain and scope of the project
- Assist the requirement engineers to make progress in the elicitation process by simulating real time stakeholder interaction

Some of the core elements and functionalities of the VSS application have been identified from the interview with the stakeholder. Although, the identified elements, their necessity, and the relationship among the elements are not concrete yet, a high-level overview of the main components of the VSS application can be provided at this point of our project. The core elements that have so far are as follows:

- **Project:** It serves as a core unit within the VSS system. The VSS can hold multiple projects at a time. Each project within the VSS system can independently derive relevant features, feedback, artifacts based on the project specific data. This unit also ensures organized data management within the VSS.
- **VirtualStakeHolder:** Represents a stakeholder with whom the requirements engineers interact. This unit simulates real alike stakeholder interactions allowing users of VSS to proceed with requirements elicitation process without the need for waiting for a human stakeholder.

- **User:** This element is the main users of the VSS system. This refers to the individuals such as requirement engineers, business analysts, Project managers who directly interact with the VSS and initiates stakeholder simulation to generate necessary information, requirements, feedback, insights based on the project domain.
- **Artifact:** Represents various requirement engineering artifacts such as user stories, personas, use case scenarios along with stakeholder feedback and insights. These are generated and managed within each projects.
- **SecurityManager:** This element responsible for data security. This segment can insures various security measures such as encryption, access control management, regulatory compliances checking.
- **AuditLog:** This segment is responsible for tracking the user interaction with the VSS system. It logs any changes made by user. Basically, this segment enhances the accountability and transparency by documenting the actions performed on project data by any user.
- **IntegrationAPI:**This segment provides an interface for connecting with various third party tools and APIs such as Github, jira and so on.
- **Response:** Represents the simulated response provided by the VSS system based on domain specific knowledge it gains over the course of the project. Responses can be stored and referenced for future use. It also maintains the consistency of the VSS stakeholder simulation among various sessions.

The high level UML class diagram for the VSS application is as follows:



3. Initial Problem Description

3.1. Stakeholders and Their Goals

3.1.1. At least 5 important stakeholders and their goals

This section presents 5 majors stakeholders and their goals. The number of goals varies from one stakeholder to another.

- **Requirement engineers:**
  - Goal 1: Facilitate requirements elicitation without the need for stakeholders physical presence
  - Goal 2: Provide realistic stakeholder answers to maintain the project in the same line as real world needs
- **Business Analyst:**
  - Goal 3: Access accurate and domain-specific insights for unfamiliar fields to guide project specifications.
  - Goal 4:Utilize the VSS to validate preliminary requirements and ensure coherence with project goals.
- **Project Manager:**
  - Goal 5:Ensure that requirements elicitation progresses without delays due to absent stakeholders.
  - Goal 6:Monitor and track requirements changes and validate that generated artifacts meet the project's objectives.
- **End Users:**
  - Goal 7:Provide input on system functionality and usability requirements through simulated feedback.
  - Goal 8:Validate that system requirements align with the end users' actual needs and expectations.
- **IT support and maintenance team:**
  - Goal 9:Maintain system security and ensure data privacy according to regulations like General Data Protection Regulation (GDPR).
  - Goal 10:Ensure system availability and troubleshoot any issues in real-time for continuous operation.
  - Goal 11:Facilitate integration of VSS with existing tools like JIRA for seamless project management.
  - Goal 12:Handle version control for artifacts, ensuring traceability and compliance with organizational standards.

3.2. Scope of the System

3.2.1. General features within the scope of the system, and explicit description of what is not in the scope of the system

In scope features:

- **Stakeholder Simulation**: The VSS will simulate stakeholder interactions, providing realistic responses based on contextual data
- **Artifact Generation**:The system will create user stories, personas, and other engineering artifacts as needed.
- **Requirements Validation**:Provides partial validation of requirements or user stories, helping ensure that the project progresses in alignment with stakeholder needs.
- **Multilingual Support**: Allows stakeholders from different regions to interact with the system in their preferred language.

Out of scope features:

- **Emotional and Intent Interpretation**:The system will not interpret complex emotions or intentions, as it relies on text-based LLMs.
- **Real-Time Human Interaction Replacement**: The VSS aims to support, not replace, nuanced interactions, particularly in complex negotiations or interpersonal communications.
- **Full Compliance with All Possible Privacy Standards**:The system focuses on GDPR, but complete compliance with all global privacy standards is out of scope.

3.2.2. Description of dependencies and anticipated interactions with external systems

- **JIRA Integration**:Facilitates smooth transitions between requirements documentation and project management activities.
- **APIs for External Integration**:Enables external tools to interface with the VSS, enhancing functionality and making it adaptable to various organizational environments.
- **Data Security and Compliance Tools**:Ensures secure data handling and compliance with relevant privacy laws, primarily GDPR, for data protection during interaction and storage.

3.3. Main User Stories

User Story 1: Elicit Requirements from Virtual Stakeholders

Identifier: US001
As a requirements engineer, I want to interact with virtual stakeholders, so that I can gather requirements even when real stakeholders are not available.

User Story 2: Generate Domain-Specific Information

Identifier: US002
As a business analyst, I want to request domain-specific information from the VSS, so that I can understand unfamiliar domains better.

User Story 3: Create Personas Automatically

Identifier: US003
As a requirements engineer, I want to generate personas automatically, so that I can have representative user profiles for the system.

User Story 4: Validate Requirements

Identifier: US004
As a business analyst, I want to validate requirements with the VSS, so that I can ensure the requirements are coherent and complete.

User Story 5: Generate User Stories

Identifier: US005
As a product owner, I want to use the VSS to create user stories, so that I can have a clear and structured list of features for development.

User Story 6: Produce Non-Functional Requirements

Identifier: US006
As a requirements engineer, I want to generate non-functional requirements using the VSS, so that I can cover aspects like performance, security, and usability.

User Story 7: Simulate Stakeholder Meetings

Identifier: US007
As a project manager, I want to simulate stakeholder meetings with the VSS, so that I can test and refine my elicitation strategies.

User Story 8: Handle Ambiguities

Identifier: US008
As a requirements engineer, I want to use the VSS to handle ambiguities in requirements, so that I can clarify and refine unclear requirements.

User Story 9: Generate Use Case Scenarios

Identifier: US009
As a business analyst, I want to create use case scenarios using the VSS, so that I can have a detailed description of system interactions.

User Story 10 (Negative User Story): Handle Inaccurate Data

Identifier: US010
As a requirements engineer, I want to recognize and flag potentially inaccurate or biased data provided by the VSS, so that I can avoid incorporating misleading information into the project requirements.

3.4. Preliminary Requirements

Updated table of preliminary requirements following the interview and other sources of information
F = Functional Requirement, NF = Non-Functional Requirement

ID	Requirement	Source	Priority	Type	Goal Addressed	User Story Addressed
1	The system shall process domain-specific data securely	External	High	NF	G3	US002
2	The system shall simulate realistic stakeholder responses based on project context data.	External	High	F	G2	US001
3	The system must maintain high availability with an uptime of 99.999%.	Internal	High	NF (Perfor-mance)		
4	The system should offer multilingual support to cater to global teams	External	Low	F	G7	US007
5	The system shall generate non-functional requirements based on context data given.	External	High	F	G2	US006
6	The system shall generate user stories based on the stakeholder's input.	Internal	High	F	G4	US005
7	The system must be accessible via a web interface compatible with modern browsers (e.g., Chrome, Firefox).	Internal	Medium	NF (Acces sibility)		
8	The system shall store previous interactions for future sessions, allowing for continuous stakeholder simulation.	Internal	Medium	F	G5	US007
9	The system shall provide suggestions for additional functional requirements based on stakeholder feedback.	Internal	Low	F		
10	The system must encrypt all sensitive data in transit and at rest.	Internal	High	NF (Securit y)	G6	US001
11	The system shall allow	External	Low	F		US001

	customization of profiles of virtual stakeholders for project specific needs					
12	The system shall allow multiple teams to access and interact with it concurrently	External	High	NF (concur rency)		
13	The system shall store generated artefacts (i.e.user stories, personas, use case scenarios) in an acceessible repository for future reference with options to retrieve, update, and delete as needed	External	High	F	G6	005
14	The system shall have version control for generated artefacts	External	Medium	NF (Mainta -inabilit- y)	G12	
15	The system shall suggest test case scenarios based on requirements	External	Low	F		
16	The VSS shall be trained with large data models in order to detect and flag potentially inaccurate or biased data in the requirements.	Internal	Medium	NF (Data Integrit- y)		US010
17	The system shall be able to handle projects of different sizes	Internal	High	NF (Scalab ility)		
18	The system shall have a user-friendly interface that enables user to perform tasks with minimal training, ensuring ease of navigation and accessibility	External	High	NF (Usabili ty)		
19	The system shall allow for real-time feedback and interaction with virtual stakeholders during elicitation sessions.	External	High	F		
20	The system shall integrate with JIRA, a popular project management tool, for seamless transition of requirements and artefacts.	External	Medium	F		
21	The system shall have an audit log feature to track changes and interactions for accountability and transparency	Internal	Medium	NF	G6	
22	The system must ensure	Internal	High	NF		

	compliance with the GDPR data protection regulations for handling user data.			(Security/Compliance)		
23	The system shall allow API access for external software application integration	External	Low	NF (Portability)		
24	The system shall be able to generate domain specific personas based on user defined context and data	External	Medium	F		US003
25	The system shall ensure that a user can generate and access any requirements artefact with no more than 5 clicks	External	Medium	NF (Usability)		
26	The system shall allow business analysts to validate preliminary requirements to ensure coherence with project goals.	Internal	Medium	F	G4	US004
27	The system shall generate non-functional requirements covering aspects such as performance, security, and usability.	Internal	High	F	G2	US006
28	The VSS shall simulate stakeholder meetings, allowing project managers to test and refine elicitation strategies.	Internal	High	F	G5	US007
29	The system shall handle ambiguities by allowing requirements engineers to refine unclear requirements.	Internal	Low	F	G1	US008
30	The system shall allow business analysts to edit and save virtual stakeholder responses as artifacts.	Internal	High	F	G4	US004
31	The VSS shall generate a summary report of elicited requirements at the end of each simulated session.	Internal	Medium	F	G5	US001
32	The VSS shall allow requirements engineers to categorize requirements by priority and urgency.	Internal	Medium	F	G2	US004
33	The VSS shall allow real-time feedback collection from end users regarding system functionality.	Internal	Low	F	G7	US007
34	The system shall maintain a repository of domain-specific	Internal	Medium	F	G3	US002

This table presents the changes that were made from deliverable 1.

	Requirements		
	Added	Removed	Modified
ID(s)	26, 27, 28, 29, 30, 31, 32, 33, 34, 35	No Requirements from Deliverable 1 were removed	12, 14, 23, 16
Explanation/Reasoning	Added 10 more requirements based on goals and User stories provided.		12 - Changed from functional to Non-functional requirement (concurrency) 14 - Changed from functional to Non-functional requirement (Maintainability) 23 - Changed from functional to Non-functional requirement (portability) 16 - Added detail on tech used to provide bias-free response

3.5. Major Risks and Their Potential Consequences

Risk 1: Limited Understanding of Specific Contexts

Description: The VSS may struggle to accurately understand the precise context of a project, a particular stakeholder, and their actual needs. Consequences:

- Inaccurate Requirements: Misinterpretation of stakeholder needs can lead to incorrect or incomplete requirements.
- Increased Rework: The development team may need to spend additional time and resources correcting these inaccuracies.
- Stakeholder Dissatisfaction: Stakeholders may be dissatisfied with the outcomes if their actual needs are not met.

Risk 2: Hallucinations in Responses

Description: LLMs often provide convincing but incorrect answers, leading to hallucinations in the responses generated by the VSS. Consequences:

- Misleading Information: Decisions based on hallucinated information can lead to flawed system design and functionality.
- Time Wastage: Engineers may waste time verifying and correcting incorrect information.
- Erosion of Trust: Continued inaccuracies can erode trust in the VSS, reducing its effectiveness and utility.

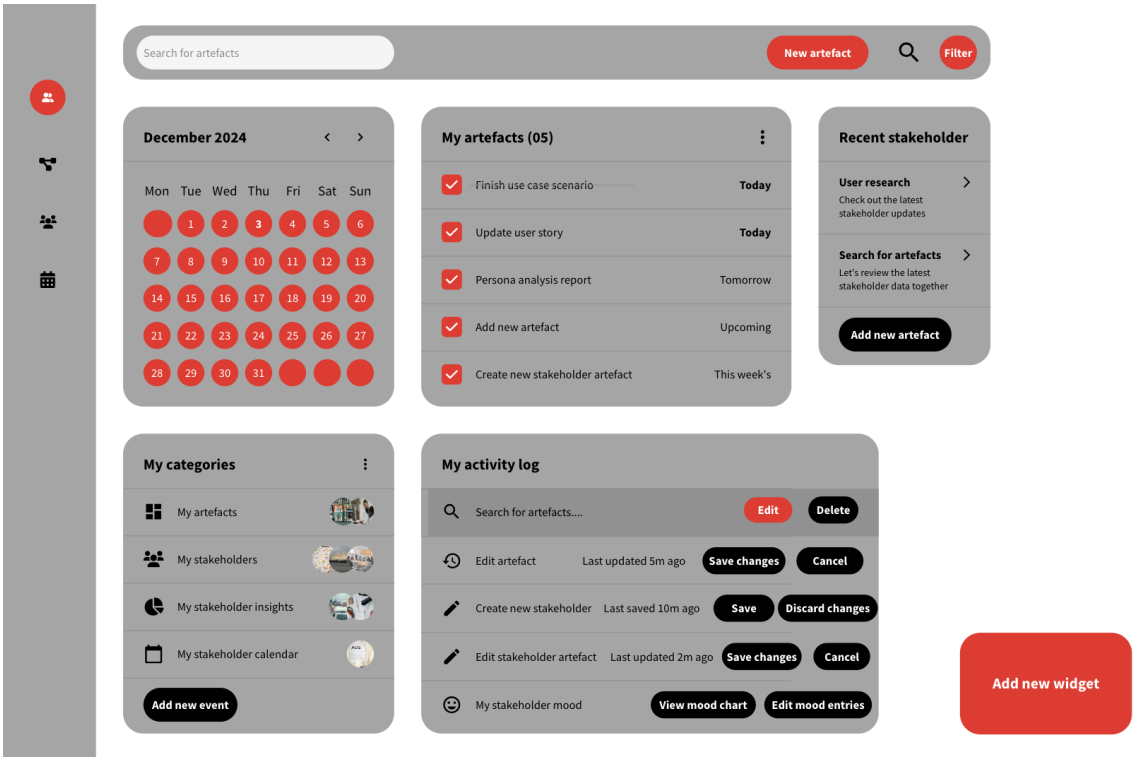
Risk 3: Data Privacy and Security Concerns

Description: The VSS may require access to sensitive or proprietary information, raising concerns about data privacy and security. Consequences:

- Data Breaches: Unauthorized access to sensitive data can lead to data breaches, legal consequences, and financial losses.
- Compliance Issues: Failure to comply with data protection regulations can result in fines and reputational damage.
- Stakeholder Reluctance: Stakeholders may be reluctant to share necessary information, hindering the effectiveness of the VSS.

4. Interface Prototypes

Screen 1: Home Screen Dashboard



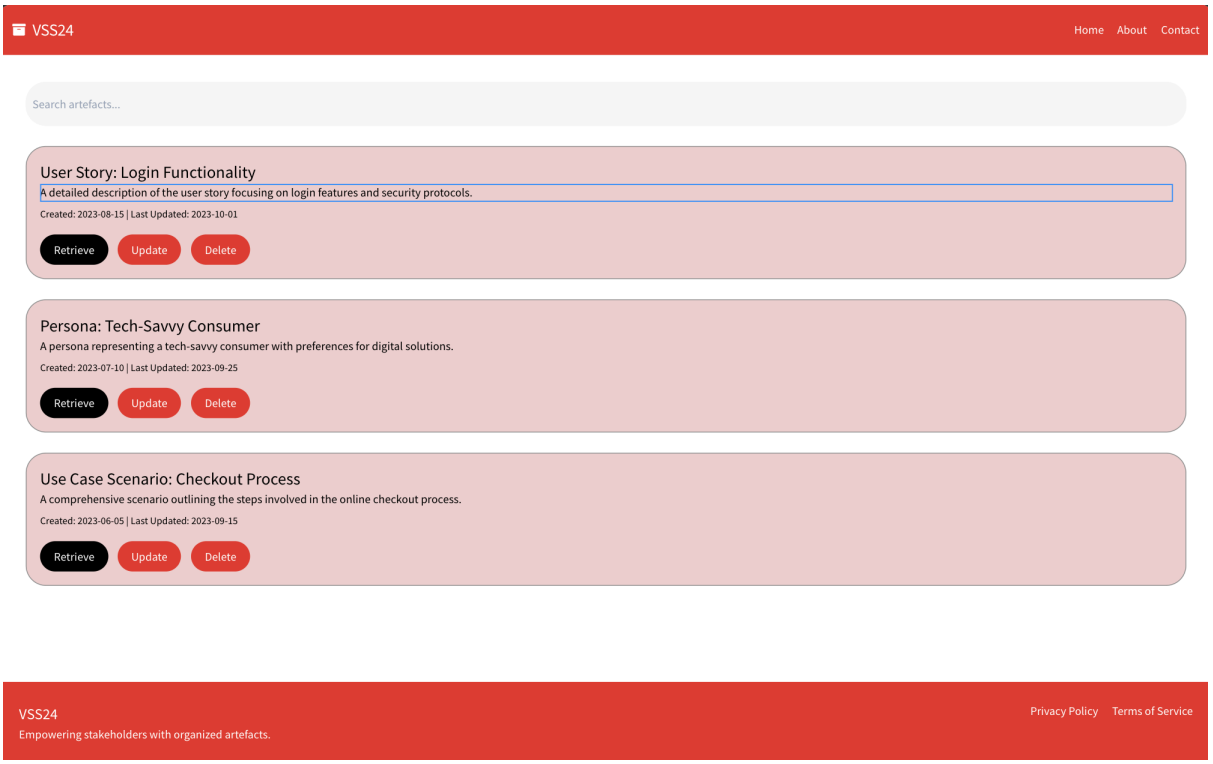
Targeted Stakeholder Goals:

- Goal 1: Facilitate requirements elicitation without the need for stakeholders physical presence
- Goal 6: Monitor and track requirements changes and validate that generated artifacts meet the project's objectives..

Features:

- Calendar Integration: Shows a monthly calendar to track important dates and deadlines.
- My Artefacts Section: Lists the user's artefacts with statuses (e.g., Today, Tomorrow, Upcoming) for tracking progress.
- Recent Stakeholder Activities: Displays recent activities related to stakeholders, aiding in monitoring stakeholder involvement.
- My Categories Section: Provides quick access to different artefacts and stakeholder-related sections.
- My Activity Log: Tracks and logs user activities related to artefact management.
- Add New Widgets and Artefacts: Allows users to add new artefacts or widgets to customize their dashboard.

Screen 2: Artifacts Management Interface



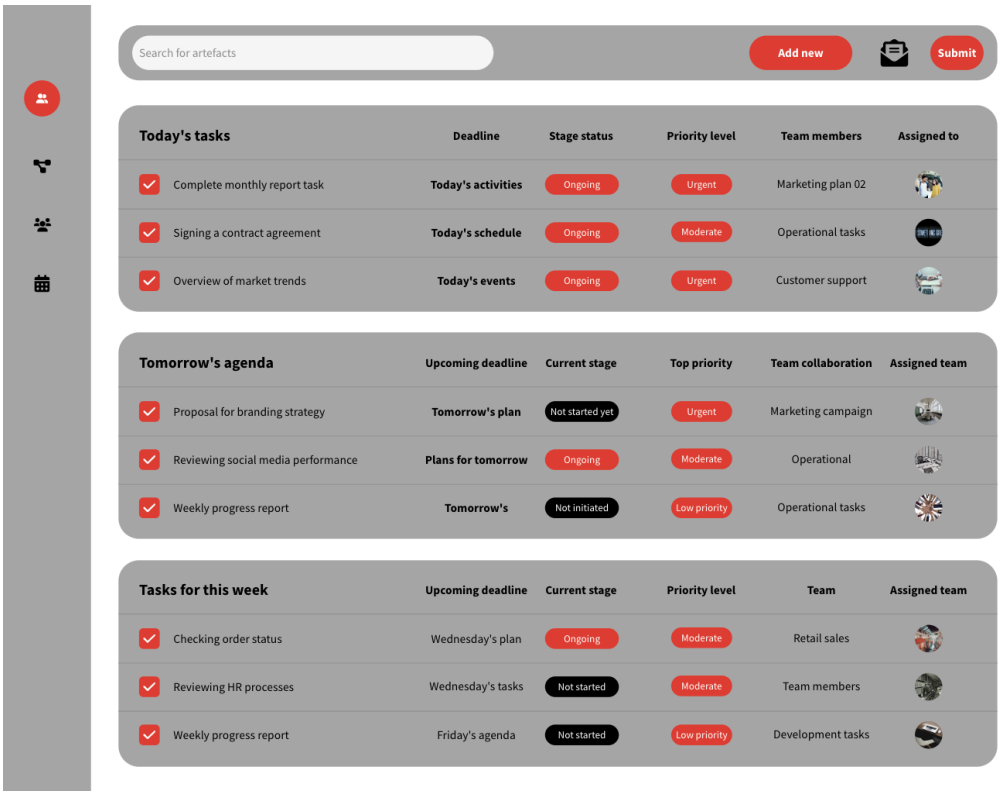
Targeted Stakeholder Goals:

- Goal 12: Handle version control for artifacts, ensuring traceability and compliance with organizational standards.

Features:

- Search Bar for Artefacts: Facilitates quick searching of artefacts.
- List of Artefacts: Displays artefacts with their descriptions, creation, and last updated dates.
- Action Buttons (Retrieve, Update, Delete): Provides options to retrieve, update, or delete artefacts, supporting artefact management and version control.
- Artefacts Metadata: Shows detailed information about each artefact, including the creation date and last update.

Screen 3: Task and Agenda Management Interface



Targeted Stakeholder Goals:

- Goal 1: Facilitate requirements elicitation without the need for stakeholders' physical presence.
- Goal 5: Ensure that requirements elicitation progresses without delays due to absent stakeholders.

Features:

- Today's Tasks: Lists tasks for the current day with deadlines, statuses, and priority levels, ensuring immediate attention to urgent tasks.
- Tomorrow's Agenda: Prepares users for the next day's tasks, helping to prioritize and plan ahead.
- Weekly Tasks: Summarizes tasks for the week to maintain an overview of ongoing activities and deadlines.
- Add New Task Button: Allows users to add new tasks directly from the interface.
- Search for Artefacts: Helps in finding specific tasks or artefacts quickly.
- Team Collaboration Section: Shows team members assigned to tasks, enhancing collaboration and clear task assignments.

5. Team Description and Roles

Here is a table summarizing the team members, their contributions, the percentage of work, and hours invested:

Team Member	Contributions	Percentage of work	Hours
Alperen Akin	Preliminary Requirements	25%	2h
Amadou Kone	-Stakeholders and Their Goals (10%) (1h) -Scope of the System (10%) (1h) -Team Description and Roles (5%) (30 min)	25%	2h30
Hazzem Sukar	-Main User Stories (10%) (30 min) -Risks (10%) (20 min) -Interface Prototypes (5%) (1h40)	25%	2h30
Shabrina Sharmin	-Domain Description (20%) (2h20) -Introduction (5%) (10 min)	25%	2h30

6. References

"GPT-4." n.d. Accessed November 8, 2024. https://openai.com/index/gpt-4-research/.

Team Member	Contributions	Percentage of work	Hours
Alperen Akin		25%	
Amadou Kone	-Played stakeholder role during validation meeting - Added requirements in HelixALM	25%	2h
Hazzem Sukar	-Lead interviewer during validation meeting - Updated requirements	25%	
Shabrina Sharmin		25%	